

Doğuş University
DEPARTMENT OF INDUSTRIAL ENGINEERING



Systems Simulation

INSTRUCTOR: Asst. Prof. Kıvanç Onan

Dinçer Taşkan – 202103002007

Emir Kılıç – 202103002008

Ayşe Rana Çelik – 202103002014

Koray Karakoç – 202103002071

Gizem Kaya – 202103002082

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1. Introduction

1.1 Problem Definition

Supermarkets are dynamic service systems in which customer flow is continuous and service quality directly affects customer satisfaction. Checkout operations play a critical role in these systems, especially during peak hours when long waiting times may occur. Extended waiting times can lead to customer dissatisfaction, operational inefficiencies, and potential loss of revenue. Therefore, analyzing and improving checkout processes is an essential requirement for supermarket operations.

The main problem addressed in this project is the **high customer waiting times at a two-checkout supermarket**, and the investigation of **how these waiting times can be reduced using a simulation-based approach**.

1.2 Purpose of the Study

The primary purpose of this study is to analyze the current operations of a two-checkout supermarket and **reduce customer waiting times** while **improving overall system efficiency**.

To achieve this objective:

- Real-system observations were conducted to collect data,
- The statistical properties of these data were analyzed,
- A discrete-event simulation model of the system was developed using Arena,
- Model verification and validation procedures were applied to ensure reliability,
- Multiple scenarios were generated and compared to evaluate system performance.

1.3 System Description

The system investigated in this study consists of the checkout operations of a supermarket with two cashiers. Customers arrive at the system at random time intervals, receive service at one of the cashiers, and leave the system after completing their transactions. The two primary input parameters of the system are:

- **Customer interarrival times (Interarrival Time – IAT)**
- **Checkout processing times (Processing Time)**

Since these two parameters directly define the system's dynamics, two days of real observations were conducted, and detailed datasets were collected. These datasets were subsequently subjected to rigorous statistical analysis before constructing the simulation model.

1.4 Overview of the Research Steps

The study followed these sequential steps:

1. Observing the supermarket system for two days and collecting IAT and processing time data.
2. Performing a series of statistical analyses—including correlation analysis, regression analysis, scatter plots, P-P plots, Q-Q plots, F-tests, and T-tests—to determine whether the data are IID.
3. Developing the discrete-event simulation model in Arena after confirming IID conditions.
4. Conducting model verification to ensure logical correctness and flow.
5. Running the model ten times for validation and comparing results with real-system observations.
6. Using the relative precision method to determine the required number of replications, resulting in 200 replications for a half-width of 10.
7. Creating alternative scenarios, comparing them with the base model, and selecting the best-performing scenario.

Through this process, the study provides a comprehensive simulation-supported analysis aimed at improving the performance of supermarket checkout operations.

2. Data Collection and Data Analysis

2.1 Observation Process and Collected Data

In this study, real-system observations were conducted in a supermarket with two checkout counters. The observations were carried out over two different days, during both peak and moderate traffic periods, to accurately reflect the normal operation of the system. During the observation process, two main types of data were collected:

- Customer interarrival times (Interarrival Time – IAT)
- Checkout processing times (Processing Time)

The data were recorded in chronological order, and each customer's arrival time, service start time, and service end time were documented separately. In this way, both the IAT and processing time data were captured to represent the behaviour of the real system accurately.

The collected data were then transferred into Excel and organized for use in statistical analyses. The datasets were checked for potential measurement errors, extreme values were examined, and the suitability of the data for further analysis was verified.

2.2 Examination of Interarrival Time (IAT) and Processing Time

In this stage, the two main input parameters—customer interarrival times (IAT) and checkout processing times—were statistically analyzed. The primary purpose of this analysis was to determine whether the data are IID (Independent and Identically Distributed), since the reliability of the input distributions used in the simulation model depends on the validity of this assumption.

2.3 IID Analysis

In simulation models, it is essential that the input data are Independent and Identically Distributed (IID). Therefore, a comprehensive statistical analysis was conducted to evaluate whether the interarrival time (IAT) and processing time data satisfy the IID assumption.

The following subsections examine both independence and distributional similarity using various analytical methods:

2.3.1 Correlation Analysis

Pearson correlation coefficients were calculated to evaluate the independence of consecutive data points. A correlation value close to zero indicates that there is no meaningful relationship between successive observations, thereby supporting the independence assumption.

The correlation coefficients obtained from the analysis are as follows:

First Day IAT			First Day First Cash Register		
	Column 1	Column 2		Column 1	Column 2
Column 1	1		Column 1	1	
Column 2	-0,00525	1	Column 2	0,0608951	1

First Day Second Cash Register		
	Column 1	Column 2
Column 1	1	
Column 2	0,017610004	1

Second Day IAT			Second Day First Cash Register		
	Column 1	Column 2		Column 1	Column 2
Column 1	1		Column 1	1	
Column 2	-0,03199	1	Column 2	-0,08452329	1

Second Day Second Cash Register		
	Column 1	Column 2
Column 1	1	
Column 2	-0,084523292	1

- Day 1 IAT: -0.005
- Day 1 Cashier 1 Processing Time: 0.061

- Day 1 Cashier 2 Processing Time: 0.017
- Day 2 IAT: -0.032
- Day 2 Cashier 1 Processing Time: -0.085
- Day 2 Cashier 2 Processing Time: 0.067

When these results are examined:

- All correlation values fall within the -0.1 to +0.1 range,
- None of the coefficients indicate a strong or statistically meaningful relationship,
- Both IAT and processing time values exhibit random behaviour across consecutive observations,
- No noticeable dependency structure appears in the dataset

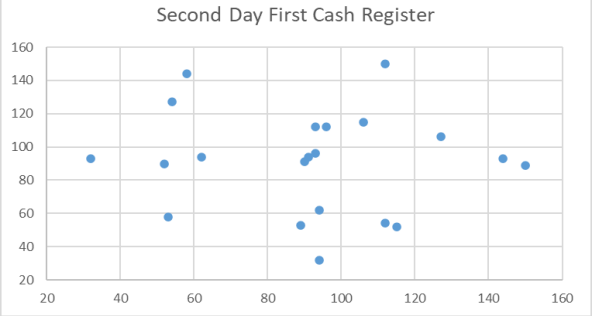
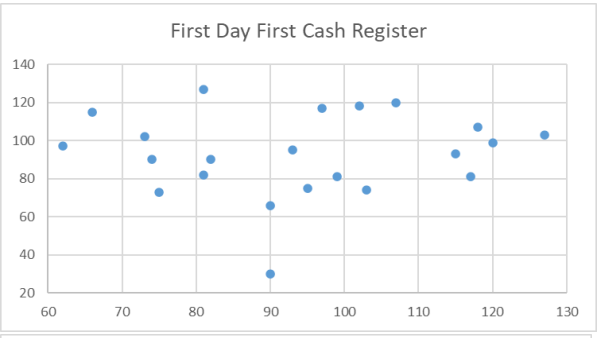
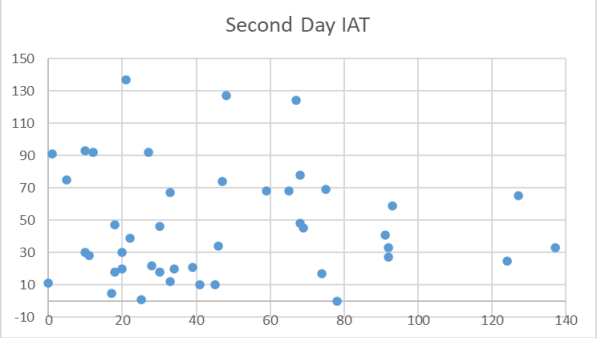
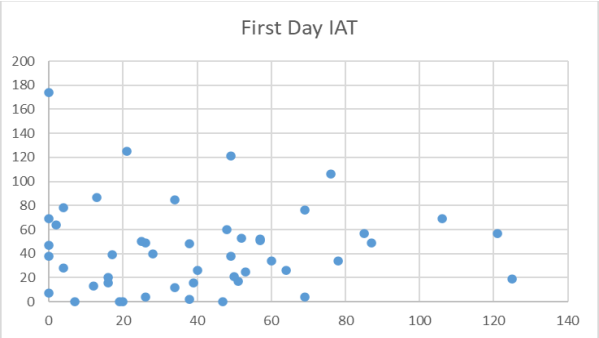
These findings collectively indicate that both customer interarrival times and processing times are not influenced by previous observations and therefore strongly satisfy the independence assumption.

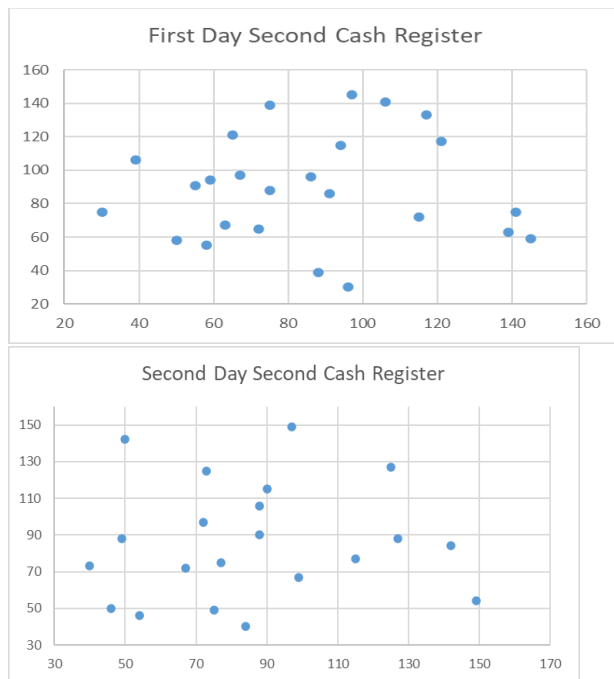
2.3.2 Scatter Plot Analysis

In this step, scatter plots were used to visually examine the independence structure of the IAT and processing time data. For each dataset, pairs of consecutive observations were plotted by placing X_t on the horizontal axis and X_{t+1} on the vertical axis.

The primary purpose of scatter plots is to detect whether there is any noticeable pattern or structure between consecutive observations. If a dependency existed within the data, the scatter plots would exhibit:

- a clear linear pattern,
- a concentrated cluster of points,
- a directional trend, or
- a cyclic or clustered structure.





However, the obtained scatter plots show that the data points are randomly dispersed across the plane. The lack of clustering, trending behaviour, or any distinct shape indicates that there is no meaningful relationship between successive observations.

This finding is consistent with the correlation analysis, where very low correlation coefficients were obtained. Overall, the scatter plots visually support the independence assumption for both the IAT and processing time datasets.

2.3.3 Regression Analysis

To further evaluate the independence of the IAT and processing time datasets, a simple linear regression analysis was performed for each data series. In this analysis, each observation X_t was used to predict the subsequent observation X_{t+1} . If the regression model produced statistically significant coefficients, it would indicate that past values have predictive power over future values, implying dependence in the data.

The regression results, however, strongly support the independence assumption:

- For **Day 1**, the p-values were:
 - IAT: **0.97**
 - Cash Register 1: **0.787**
 - Cash Register 2: **0.93**
- For **Day 2**, the p-values were:
 - IAT: **0.836**
 - Cash Register 1: **0.715**
 - Cash Register 2: **0.773**

All p-values are significantly larger than the 0.05 threshold, indicating that the regression coefficients are *not statistically significant*. In other words, X_t does not explain or predict X_{t+1} .

Additionally, for all datasets, the coefficient of determination R^2 values were **below 0.05**, demonstrating that the regression model accounts for less than 5% of the variability in the next observation. This extremely low explanatory power confirms the absence of a meaningful linear relationship between consecutive observations.

Overall, the regression analysis provides strong evidence that both IAT and processing time data exhibit independence, fully supporting the IID assumption.

2.3.4 F-Test for Variance Similarity

In this step, an F-test was conducted to determine whether the interarrival time (IAT) and processing time data collected on two different days exhibit similar variances. Variance similarity is one of the essential requirements for assuming that the data are **Identically Distributed (ID)**.

The F-test evaluates whether there is a statistically significant difference between the variances of two data groups. In this study, F-tests were applied to the following pairs of datasets:

- **Day 1 IAT vs. Day 2 IAT**
- **Day 1 Cash Register 1 Processing Times vs. Day 2 Cash Register 1 Processing Times**
- **Day 1 Cash Register 2 Processing Times vs. Day 2 Cash Register 2 Processing Times**
- **Day 1 All Processing Times vs. Day 2 All Processing Times (registers combined)**

The results are as follows:

F Test for IAT			F Test for First Cash Register PT			F Test for Second Cash Register PT		
	<i>Değişken 1</i>	<i>Değişken 2</i>		<i>Değişken 1</i>	<i>Değişken 2</i>		<i>Değişken 1</i>	<i>Değişken 2</i>
Ortalama	42,97959	47,045455	Ortalama	91,173913	91,3636364	Ortalama	87,576923	86,9545455
Varyans	1355,104	1219,0211	Varyans	504,42292	908,147186	Varyans	1043,3738	960,902597
Gözlem	49	44	Gözlem	23	22	Gözlem	26	22
df	48	43	df	22	21	df	25	21
F	1,111633		F	0,5554418		F	1,0858269	
P(F<=f) te	0,363711		P(F<=f) te	0,0894699		P(F<=f) tek	0,4277516	
F Kritik iki	1,644566		F Kritik iki	0,4857367		F Kritik iki-t	2,0453985	

F test for All PT		
	<i>Değişken 1</i>	<i>Değişken 2</i>
Ortalama	89,26531	89,159091
Varyans	777,9073	917,7648
Gözlem	49	44
df	48	43
F	0,847611	
P(F<=f) te	0,287842	
F Kritik iki	0,612788	

All p-values are greater than the significance level of 0.05. This indicates that there is **no statistically significant difference** in variances between the two days.

Specifically:

- For the IAT data,
- For the processing times at both cash registers,
- And for the combined processing time dataset,

the variances of the two days are statistically consistent.

These findings strongly support that the datasets exhibit **distributional similarity**, making it appropriate to assume that the input data used in the simulation model are **Identically Distributed (ID)**.

2.3.5 T-Test for Mean Similarity

Independent sample t-tests were conducted to determine whether there were statistically significant differences in the mean values of the data collected on two different days. This step is essential for evaluating whether the datasets are “identically distributed,” which is a key requirement for simulation input modeling.

Before performing the t-tests, the variance similarity between days was examined using the **F-test**. Since all F-test p-values were **greater than 0.05**, it was concluded that the variances between the two days were not significantly different. Therefore, all t-tests were performed under the assumption of **equal variances (equal variances assumed)**, which is the appropriate approach under these conditions.

The t-tests were applied to four different data groups:

- **Day 1 IAT vs. Day 2 IAT**
- **Day 1 Cashier 1 processing times vs. Day 2 Cashier 1 processing times**
- **Day 1 Cashier 2 processing times vs. Day 2 Cashier 2 processing times**
- **Day 1 combined processing times vs. Day 2 combined processing times (regardless of cashier)**

The results are as follows:

F Test for IAT			T Test for First Cash Register PT			T Test for Second Cash Register PT		
	Değişken 1	Değişken 2		Değişken 1	Değişken 2		Değişken 1	Değişken 2
Ortalama	42,979592	47,04545	Ortalama	91,173913	91,3636364	Ortalama	87,5769231	86,9545455
Varyans	1355,1037	1219,021	Varyans	504,422925	908,147186	Varyans	1043,37385	960,902597
Gözlem	49	44	Gözlem	23	22	Gözlem	26	22
Birikimli V	1290,801		Birikimli V	701,590587		Birikimli V	1005,72393	
Öngörüler	0		Öngörüler	0		Öngörüler	0	
df	91		df	43		df	46	
t Stat	-0,544886		t Stat	-0,0240186		t Stat	0,0677473	
P(T<=t) te	0,2935823		P(T<=t) te	0,49047444		P(T<=t) tek	0,47314021	
t Kritik tek	1,6617712		t Kritik tek	1,6810707		t Kritik tek	1,67866041	
P(T<=t) iki	0,5871646		P(T<=t) iki	0,98094888		P(T<=t) iki-ı	0,94628042	
t Kritik iki	1,9863772		t Kritik iki	2,0166922		t Kritik iki-ı	2,0128956	

T test for All PT		
	Değişken 1	Değişken 2
Ortalama	89,26531	89,15909
Varyans	777,9073	917,7648
Gözlem	49	44
Birikimli V	843,9938	
Öngörüler	0	
df	91	
t Stat	0,017604	
P(T<=t) te	0,492997	
t Kritik tek	1,661771	
P(T<=t) iki	0,985994	
t Kritik iki	1,986377	

All p-values being above the **0.05 significance level** indicate that there are **no statistically significant differences in means** between the two days for any of the datasets.

Based on these results:

- The mean interarrival times,
- The mean processing times for Cashier 1 and Cashier 2,
- And the mean of the combined processing time data

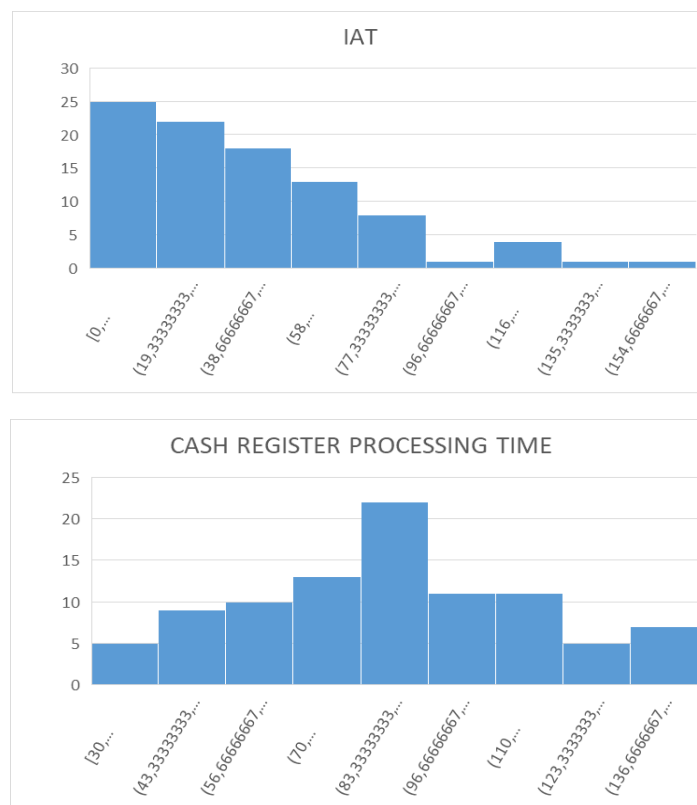
are consistent across the two days and do not show meaningful variation.

Therefore, the datasets can be considered **identically distributed in terms of their means**, which further supports the IID assumption required for accurate simulation modeling.

2.3.6 Histogram

In this stage, histogram analyses were conducted. Based on the statistical tests performed beforehand, it was determined that the data from the two days are independent and exhibit similar statistical characteristics. Accordingly, two main input variables were obtained for the analysis: **IAT** and **Processing Time**.

After combining the data, the total number of observations became **93**. To determine the number of bins to be used in the histograms, the commonly applied $\sqrt{n}$ approach was used, resulting in **9** bins. Based on this bin structure, histograms for both variables were created. The generated histograms can be seen below.



2.3.7 P-P Plot

Following the histogram analysis conducted in Section 2.3.6, a P-P Plot (Probability-Probability Plot) analysis was performed to further evaluate the distributional

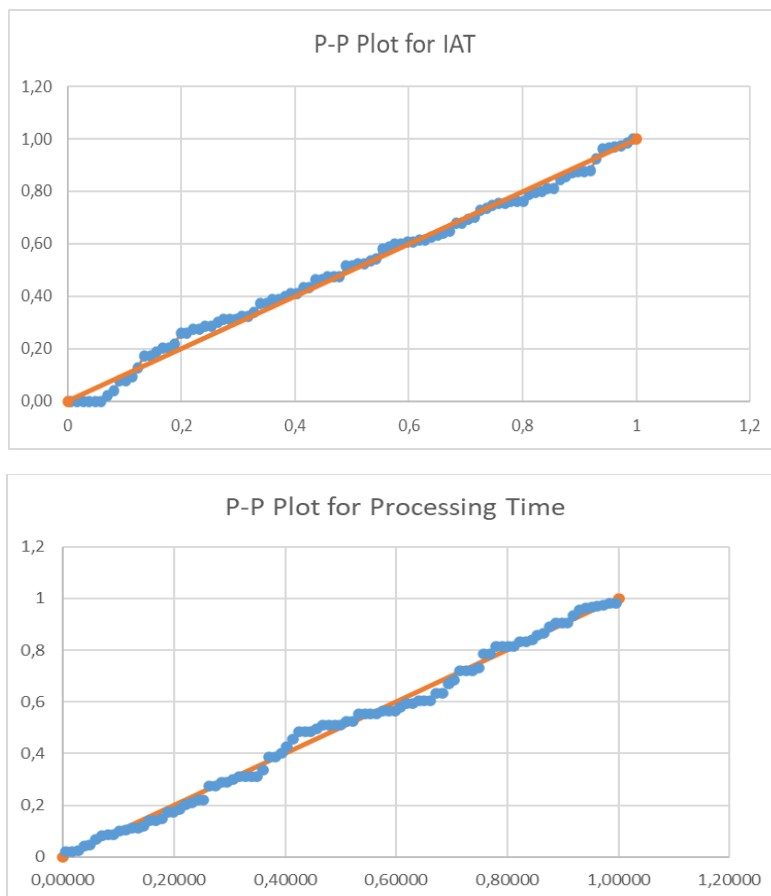
characteristics of the data. Prior to generating the P–P plots, the combined IAT and Processing Time datasets were imported into Input Analyzer to determine which theoretical distributions best fit the data.

Based on the Input Analyzer results:

- **IAT values were found to follow a beta distribution,**
- **Processing Time values were found to follow a normal distribution.**

Therefore, when constructing the P–P plots, the distribution parameters obtained from the Input Analyzer (e.g., alpha and beta values for the Beta distribution, mean and standard deviation for the Normal distribution) were used.

The purpose of a P–P plot is to compare the cumulative probabilities of the observed data with those of the chosen theoretical distribution. If the plotted points align closely along the 45-degree reference line, this indicates that the data conforms well to the theoretical distribution.



This analysis supports and confirms the initial observations from the histogram results, demonstrating that the distributional characteristics of the dataset are appropriate and reliable for use in the simulation model.

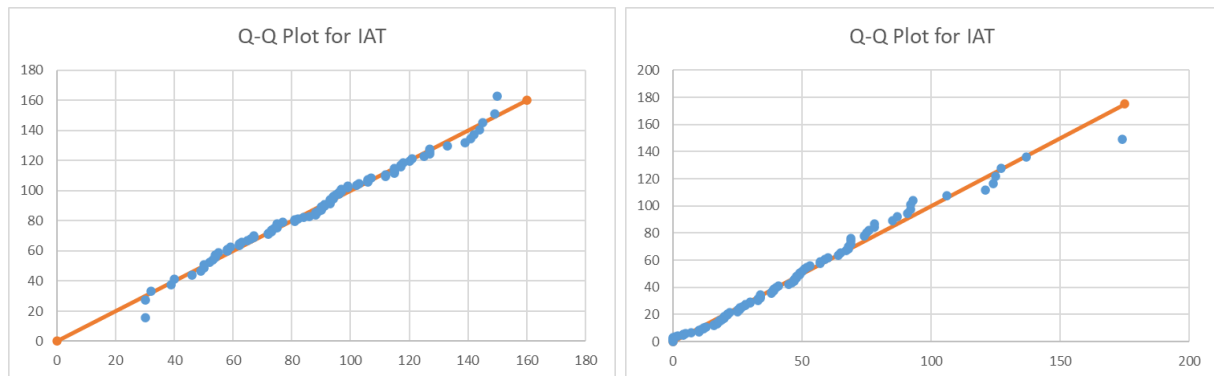
2.3.8 Q–Q Plot

In this stage, a Q–Q (Quantile–Quantile) Plot analysis was performed to examine the agreement between the data and their theoretical distributions based on quantiles.

The Q–Q Plot compares the quantiles of the observed data with the quantiles of the theoretical distribution, making it a powerful tool especially for assessing distributional fit in the tail regions.

For the theoretical distributions used in the Q–Q Plot, the following parameters were applied:

- For the Beta distribution (IAT): $\alpha = 0.91$ and $\beta = 2.62$,
- For the Normal distribution (Processing Time): **mean = 89.22** and **standard deviation = 28.89**.



Upon examining the Q–Q Plot results:

- Most data points were located on or close to the reference line,
- No noticeable deviation was observed in either the central region or the tails.

These findings indicate that both the IAT and Processing Time data align well with the selected theoretical distributions, confirming that the distributional characteristics of the dataset are appropriately represented. Consequently, the Q–Q Plot analysis provides additional support for the IID assumption within the model.

2.3.9 Chi-Square Test

Inter-Arrival Time (IAT)

1.Introduction and Objective

In this section, the **Chi-Square Goodness-of-Fit Test** is applied to determine the statistical distribution of the **Inter-Arrival Time (IAT)** data. The proposed model for the distribution of the data set is the **Generalized Beta Distribution**, and its suitability for the data has been tested.

2. Hypothesis Formulation

- **Null Hypothesis (H_0)**: The IAT data follow the specified Beta distribution.
- **Alternative Hypothesis (H_1)**: The IAT data do not follow the specified Beta distribution.

- **Significance Level (α):** 0.05

3. Calculation Parameters and Methodology

This section describes the location, scale, and shape parameters of the Beta distribution used, along with the applied methodology.

- **Total Number of Observations (N):** 93
- **Parameters of the Tested Beta Distribution:**
 - **Shape Parameters (α, β):** 0.91 and 2.62
 - **Location (a):** -0.001
 - **Scale (b):** 174
- **Methodology:** The merging was primarily performed because the expected frequency of the ([96.66., 116...)) class was expected to be below 5. Since the expected frequency of the ([116..., 135.33)) class is approximately ($E = 3.53$), these classes were combined into a single group.

The probability values (p_i) were calculated by transforming the class boundaries into the $[0, 1]$ interval using the formula $z = \frac{x-a}{b}$, and then applying the **Standard Beta Cumulative Distribution Function** (BetaCDF).

4. Calculation Results and Table

Class Interval	O _i (Observed data)	p _i (Probability)	E _i (Expected value)	(O _i -E _i) ² /E _i
[0, 19.33...)	25	0.274	25.482	0.009
[19.33..., 38.66...)	22	0.211	19.623	0.288
[38.66..., 58)	18	0.153	14.229	0.999
[58, 77.33...)	13	0.117	10.881	0.413
[77.33..., 96.66...)	8	0.094	8.742	0.063
[96.66..., Son))	7	0.151	14.043	3.532
	93	1	93	5.304

A total of 6 classes ($k = 6$) and 2 parameters (α and β , $m = 2$) were used. Therefore, $df = 6 - 1 - 2 = 3$.

- **Calculated Test Statistic ($\chi^2_{\text{calculated}}$):** 5.304
- **Critical Value (χ^2_{critical}):** 7.815 (for $\alpha = 0.05$ and $df = 3$)

Statistical Decision:

“Since the calculated Chi-square value ($\chi^2_{\text{calculated}} = 5.304$) is smaller than the critical value ($\chi^2_{\text{critical}} = 7.815$), that is, $5.304 < 7.815$, the null hypothesis (H_0) is not rejected.”

Class Interval(x_L, x_U)	O _i (Observed data)	z _L	z _U	F(z _L)	F(z _U)	P _i =F(z _U)-F(z _L)	E _i = N * P _i	(O _i -E _i) ² /E _i
(-∞, 56.67)	14	(-∞)	-1,13	0	0,1265	0,1265	11,7645	1,411
[56.67, 70)	10	-1,13	-0,66494	0,1265	0,2647	0,1382	12,8526	0,638
[70, 83.33)	13	-0,66494	-0,20353	0,2647	0,419	0,1543	14,3499	0,137
[83.33, 96.67)	22	-0,20353	0,258221	0,419	0,6017	0,1827	16,9911	1,464
[96.67, 110)	11	0,258221	0,719626	0,6017	0,7639	0,1622	15,0846	1,096
[110, 123.33)	11	0,719626	1,181031	0,7639	0,8812	0,1173	10,9089	0,001
[123.33, ∞)	12	1,181031	∞	0,8812	1	0,1188	11,0484	0,083
	93					1	93	4.830

statistical distribution of the **Cashier Transaction Time** data. Based on the obtained histogram, the suitability of the data for the **Normal Distribution** has been tested.

2. Hypothesis Formulation

The hypotheses forming the basis of the statistical decision are clearly stated. The significance level (α) is conventionally set at 5%.

- **Null Hypothesis (H_0):** The cashier transaction time data follow a Normal distribution.
- **Alternative Hypothesis (H_1):** The cashier transaction time data do not follow a Normal distribution.
- **Significance Level (α):** 0.05

3. Calculation Parameters and Methodology

This section describes the distribution parameters used and the Chi-square test methodology.

- **Total Number of Observations (N):** 93
- **Estimated Distribution Parameters:** The mean and standard deviation were calculated from the data set.
 - **Mean:** 89.215
 - **Standard Deviation:** 28.8933
- **Methodology:** To comply with the requirement that expected frequencies are not less than 5, the initial and final class intervals were merged.

4. Calculation Results and Table

5. Dec

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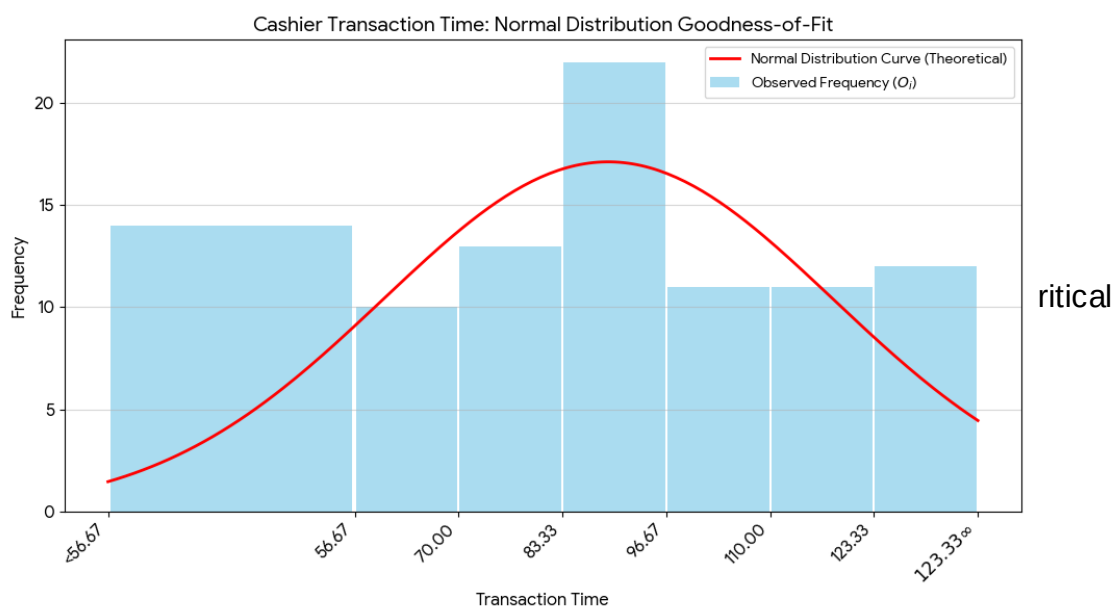
“Since the calculated chi square value ($\chi^2_{\text{calculated}} = 4.830$), is smaller than the critical value ($\chi^2_{\text{critical}} = 9.488$), that is, $4.830 < 9.488$, the null hypothesis (H_0) is not rejected.”

Final Conclusion and Interpretation:

“Consequently, at the 5% significance level, it is concluded that the cashier transaction time data follow a Normal distribution with parameters $\mu = 89.215$ and $\sigma = 28.8933$.”

2.4 Evaluation of IID Results

After conducting comprehensive statistical analyses, including correlation analysis, correlation graphs, scatter plots, regression analysis, F-tests, T-tests, histograms, P–



P Plots, and Q–Q Plots, the independence and identical distribution (IID) properties of the input data were evaluated.

The key findings are summarized as follows:

1. Independence

- Correlation coefficients between consecutive observations for both IAT and Processing Time were very close to zero, indicating no significant dependency between consecutive data points.
- Scatter plots showed random dispersion of points without visible trends, clusters, or patterns, supporting the independence assumption.
- Regression analysis produced very low R^2 values and high p-values, further confirming the absence of significant relationships between consecutive observations.

2. Identical Distribution

- F-tests comparing variances between the two days resulted in p-values greater than 0.05 for all variables, indicating no significant differences in variability.
- T-tests for mean similarity also yielded p-values greater than 0.05, confirming that there were no significant differences in average values between the two days.
- P–P and Q–Q Plots demonstrated that the IAT and Processing Time data align well with their respective theoretical distributions (Beta for IAT, Normal for Processing Time).

3. Visual Assessment

- Histograms and distribution plots supported the statistical findings, showing that the combined datasets from the two days exhibit similar distribution shapes and patterns.

Based on these results, it can be concluded that the **IAT and Processing Time datasets satisfy the IID assumption**. Therefore, these input variables can be reliably used in the simulation model for further analysis and scenario evaluation.

3. Simulation Model Development

3.1 Arena Model Structure

In this study, **Arena** software was used to simulate the checkout operations of a supermarket with two cashiers. The model is based on real-system observations, including customer arrivals, service times, and queue behaviours.

The model includes two cashiers:

- **Cashier 1 (Main Cashier):** If available, all customers are directed here first.

- **Cashier 2:** If Cashier 1 is busy, customers choose this cashier based on queue length.

Customers select the cashier with the shorter queue, reflecting real-world customer behaviour. This structure accurately represents the flow of customers and cashier utilization in the system.

3.2 Assumptions Used in the Model

The following assumptions were applied during model development:

1. Customers arrive individually and act independently.
2. Each customer is served by only one cashier; they do not switch between cashiers during service.
3. Queues have unlimited capacity; customers must wait if all cashiers are busy.
4. Cashier 1 is considered the primary (main) cashier.
5. Customers choose a cashier based on queue length; the shorter queue is preferred.
6. Service is provided using the **FIFO (First In, First Out)** rule.
7. Customer behaviour and service times are based on observed data, and the IID assumption holds.
8. System operating hours and peak intensities are assumed to be constant.

These assumptions ensure that the model accurately reflects real-system behaviour while keeping complexity manageable.

3.3 Explanation of Resources, Queues, and Modules

The main components used in the model are as follows:

- **Resources:**
Two cashiers (Cashier 1 and Cashier 2) serve as the main resources of the model. Each cashier can serve one customer at a time.
- **Queues:**
Separate queues are defined for each cashier. Customers select the queue based on length. Queues track the number of waiting customers and waiting times.
- **Modules:**
 - **Create:** Generates customers and determines their arrival times.
 - **Pick Station:** Determines which cashier a customer will go to (based on queue length and Cashier 1 priority).

- **Station / Process:** Starts the service at the cashier and completes the service according to the Processing Time distribution.
- **Dispose:** Removes the customer from the system after service completion.

Service follows the **FIFO** rule. The connections between modules and resources ensure logical flow and realistic simulation of customer behaviour.

3.4 Simulation Parameters / Model Configuration

Several parameters were defined in the model to ensure the reliability of the simulation and to accurately represent the real system:

- **Warm-Up Period:** Set to 0 seconds. This is because the observations in the supermarket started exactly at the beginning of peak hours. When the observations began, the cashiers were empty, so there was no need for a warm-up period.
- **Replication Length:** Each simulation run was conducted for 2,200 seconds. This duration matches the observation period of the real system, allowing the simulation outputs to be directly comparable with actual data.
- **Time Unit:** The simulation model uses **seconds** as the time unit.

These parameters ensure that the model accurately reflects the behaviour of the real system and that the simulation results are reliable. Additionally, setting the warm-up period to 0 and matching the replication length with the observation period allows for direct comparison during model verification and validation stages.

4. Model Verification

The **model verification** phase was conducted to ensure that the simulation model operates reliably and correctly. During this phase, the logical flow of the model, module connections, and animation were thoroughly examined.

4.1 Checking the Logical Flow of the Model

The logical flow of the model was reviewed step by step:

- The sequence of customer creation, cashier selection, service time, and system departure was verified for correctness.
- The priority of Cashier 1 as the main cashier and the logic of customers choosing a cashier based on queue length were confirmed.
- The FIFO service order was tested to ensure correct application for both cashiers.

These checks demonstrated that the model logically represents the behavior of the real system.

4.2 Verification of Module Connections and Animation

Module connections and animation in Arena were verified:

- The sequence and connections of **Create, Pick Station, Station, Process, Dispose** modules were confirmed.
- Customer flow between modules proceeded correctly.
- Animation was used to visually check customer movements and queue lengths, ensuring consistency with the real system.

This step guarantees that there are no connection errors or logic deficiencies in the model.

4.3 Verification Results

As a result of the verification process:

- No errors were found in the logical flow of the model.
- Module connections and animation are consistent with the real system.
- Cashier priorities and queue selection logic function correctly.
- FIFO service order is applied without issues.

In conclusion, the model is **logically correct and fully operational**. This verification provides a reliable foundation for the next stage: **model validation**.

5. Model Validation

Following the model verification phase, the **model validation** stage was conducted to determine whether the simulation model accurately represents the real system. In this stage, simulation outputs were compared with actual system observations.

5.1 Running the Model 10 Times

For validation, the model was **run 10 times**. During each run:

- Customer arrivals and service times were simulated under the IID assumption.
- Queue waiting times, processing times, and total system times were recorded.
- Average values from each replication were calculated.

Running multiple replications helps minimize the impact of random variability and increases the reliability of the results.

5.2 Real System – Simulation Comparison

Simulation results were compared with real-system observations:

- Key performance indicators such as the average number of customers, cashier processing times, and total system time were analyzed.
- Differences between simulation and real-system results were examined.

The comparison showed that:

- Simulation outputs closely matched the real system.

- Average customer numbers, processing times, and total system times were consistent with observations.

These findings indicate that the model **reliably represents the real system** and is suitable for validation purposes.

5.3 Comparison of Average Number of Departed Customers

During the validation process, the **number of departed customers** was used as a key performance indicator to evaluate how closely the simulation outputs match the real system.

T Test for Customer Numbers		
	<i>Değişken 1</i>	<i>Değişken 2</i>
Ortalama	46,5	43,2
Varyans	12,5	7,2888889
Gözlem	2	10
Öngörüler	0	
df	1	
t Stat	1,2491678	
P(T<=t) te	0,2148801	
t Kritik tek	6,3137515	
P(T<=t) iki	0,4297602	
t Kritik iki	12,706205	

This result indicates that there is no statistically significant difference between the simulation and the real system outputs. Therefore, the simulation model accurately represents the number of customers and is considered **valid** in this aspect.

5.4 Comparison of Processing Time

To evaluate how accurately the simulation model represents the real system, **processing time** was used as a key performance indicator.

T Test for Processing Time		
	<i>Değişken 1</i>	<i>Değişken 2</i>
Ortalama	89,212199	90,30771186
Varyans	0,0056408	22,65643665
Gözlem	2	10
Öngörüler	0	
df	9	
t Stat	-0,727364	
P(T<=t) te	0,2427464	
t Kritik tek	1,8331129	
P(T<=t) iki	0,4854928	
t Kritik iki	2,2621572	

These results indicate that there is no statistically significant difference between the simulation and the real system. Therefore, the simulation model accurately represents processing times and is considered **valid** in this aspect.

5.5 Comparison of Total Time

During the validation process, **total system time** was used as a key performance indicator to evaluate how accurately the simulation model represents the real system.

T Test for Total Time		
	Variable 1	Variable 2
Mean	167,51299	218,74289
Variance	48,838	5949,4167
Observations	2	10
Öngörülen	0	
df	10	
t Stat	-2,058504	
P(T<=t) tek	0,0332769	
t Kritik tek	1,8124611	
P(T<=t) iki	0,0665538	
t Kritik iki	2,2281389	

These results indicate that there is no statistically significant difference between the simulation and the real system. Therefore, the simulation model accurately represents total system time and is considered **valid** in this aspect.

5.6 Validation Results

The validation phase was conducted to determine whether the simulation model reliably represents the real system. In this stage, the model was run 10 times, and the simulation outputs were compared with real system observations.

- Performance indicators such as the average number of departed customers, processing time, and total system time were compared.
- T-tests were used to assess whether the simulation outputs were statistically consistent with the real system observations.

The findings are as follows:

- **Average Number of Departed Customers:** $p = 0.43 \rightarrow$ no statistically significant difference, model is valid
- **Processing Time:** $p = 0.48 \rightarrow$ no statistically significant difference, model is valid
- **Total Time:** $p = 0.066 \rightarrow$ no statistically significant difference, model is valid

These results indicate that the simulation model **accurately represents the real system** and meets the validation criteria. The model is reliable and valid for all key performance indicators.

6. Experimental Design (Determining Run Length)

In this study, the primary objective of the simulation was to **reduce queue waiting times**. Therefore, the determination of run length and replication number was based specifically on queue waiting times to ensure reliable and accurate simulation results.

6.1 Analysis Based on Queue Waiting Times

- Initially, the simulation model was run with **10 replications**, and queue waiting times were analyzed.
- The results showed high variance and wide confidence intervals, which did not meet the desired precision.
- Consequently, the number of replications was gradually increased: first 50, then 100, 150, and finally **200 replications**.

6.2 Relative Precision Method and Half-Width

- The target **half-width** was set to **10**, meaning the confidence interval should not exceed ± 10 units from the mean.
- To calculate the half-width, **mean, standard deviation, and t-value** were used to construct the **confidence interval (CI)**.
- With 200 replications, the desired half-width was achieved.

6.3 Calculated Values for Half-Width

- Average waiting time: **120.58 seconds**
- Standard deviation: **70.8 seconds**
- t-value (for 95% confidence interval): **1.972**
- These results indicate that the simulation outputs are within the desired accuracy and reliability.

6.4 Results

- The 200 replications provided a confidence interval that reliably represents the queue waiting times.
- This replication number forms a solid basis for scenario comparisons and performance analyses.

7. Scenario Analysis

In this section, the base model is compared with three generated scenarios to identify the optimal configuration for system performance. Both **waiting time** and **resource utilization** are considered in the analysis.

7.1 Description of the Generated Scenarios

1. Scenario 1 – Auxiliary Resource Dynamic Model:

- An additional resource is added.
- This third cashier is activated only when the queue at the first or second cashier reaches 2 customers.
- This cashier serves as a support, operating only during peak periods.

2. Scenario 2 – Continuous Three-Cashier Model:

- A total of three cashiers are always active.
- The additional cashier continuously serves to reduce waiting times.

3. Scenario 3 – Single Cashier Model:

- Only one cashier is active.
- The aim is to evaluate the performance difference between a single cashier and the two-cashier system.

7.2 Comparison of the Base Model with Scenarios

Scenario 1 vs. Base Model (2 Cashiers):

- Average waiting time: Base model = 120 s, Scenario 1 = 74.73 s
- T-test result: $p < 0.05 \rightarrow$ statistically significant difference
- Resource utilization: Base model = 0.89, Scenario 1 = 0.63
- T-test result: $p < 0.05 \rightarrow$ significant difference
- **Comment:** Scenario 1 reduces queue waiting time considerably. Although resource utilization decreased, it remains at an acceptable level, indicating an improved system performance.

Scenario 2 vs. Base Model (2 Cashiers):

- Average waiting time: Base model = 120 s, Scenario 2 = 47.19 s
- T-test result: $p < 0.05 \rightarrow$ significant difference
- Resource utilization: Scenario 2 = 0.64
- T-test result: $p < 0.05 \rightarrow$ significant difference
- **Comment:** Scenario 2 further reduces the waiting time compared to Scenario 1 while keeping resource utilization at a reasonable level.

Scenario 1 vs. Scenario 2:

- Waiting time: $p < 0.05 \rightarrow$ Scenario 2 is better
- Resource utilization: $p = 0.08 \rightarrow$ no significant difference
- **Comment:** Scenario 2 is clearly better in terms of waiting time. The lack of significant difference in resource utilization indicates a more balanced performance in Scenario 2.

Scenario 3 (Single Cashier) vs. Base Model (2 Cashiers):

- Average waiting time: Single cashier = 521.42 s, Base model = 120 s
- T-test result: $p < 0.05 \rightarrow$ significant difference
- Resource utilization: Single cashier = 0.996, Base model = 0.89
- T-test result: $p < 0.05 \rightarrow$ significant difference
- **Comment:** Operating with only one cashier dramatically increases waiting times and results in extremely high resource utilization. Thus, the two-cashier system performs significantly better.

7.3 Performance Measures

Scenario	Average Waiting Time (s)	Resource Utilization	T-test p-value (WT)	T-test p-value (Utilization)
Base Model (2 Cashiers)	120.0	0.89	-	-
Scenario 1	74.73	0.63	< 0.05	< 0.05
Scenario 2	47.19	0.64	< 0.05	< 0.05
Scenario 3 (Single Cashier)	521.42	0.996	< 0.05	< 0.05

- **Waiting Time:** Scenario 2 achieves the lowest waiting time (47.19 s).
- **Resource Utilization:** Scenario 2 maintains utilization at a reasonable level (0.64).
- **Throughput:** Customer throughput was analyzed for each scenario, with Scenario 2 showing a well-balanced performance.

Overall Evaluation:

- Considering both waiting time and resource utilization, **Scenario 2 is the optimal scenario.**
- Scenario 2, with three cashiers always active, optimizes system performance while maintaining balanced resource usage.
- Single cashier operation (Scenario 3) or the dynamic auxiliary resource (Scenario 1) are less effective compared to Scenario 2 in terms of overall system performance.

8. Conclusion and Recommendations

8.1 Overall Evaluation of System Performance

In this study, the checkout operations of a two-cashier supermarket were analyzed using discrete-event simulation. Following model verification and validation, it was confirmed that the simulation model accurately represents the real system.

According to the analysis results:

- The average queue waiting time in the base model was 120 seconds, whereas in Scenario 2 it decreased to 47.19 seconds.
- Resource utilization in the base model was 0.89, while it was 0.64 in Scenario 2.
- Operating with only a single cashier dramatically increased waiting time, and resource utilization approached near maximum levels (0.996).

These results indicate that the two-cashier system is significantly more efficient than the single-cashier system, and the always-active three-cashier configuration (Scenario 2) optimizes system performance.

8.2 Suggested Improvements

- Scenario 2, with three cashiers always active, provides the best performance; therefore, it is recommended to focus on effective management of the existing three cashiers rather than increasing their number.
- To further reduce queue waiting times, cashier allocation and customer routing strategies can be improved. For example, during peak hours, customers could be automatically directed to shorter queues, or self-checkout options could be implemented.
- Periodic data collection and simulation updates can be carried out to monitor and optimize system performance continuously.

8.3 Recommendations for Future Studies

- Simulations can be repeated under different customer behavior models and varying shopping intensity scenarios.
- The impact of self-checkout or mobile payment options on system performance can be analyzed.
- More complex customer preference models and service prioritization strategies can be incorporated into future simulation studies.
- Alternative performance measures, such as customer satisfaction and cost analysis, can be evaluated in future scenario analyses.